Industrial Internship Report on

” Customer Churn Prediction using Machine Learning”

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 4 weeks’ time.

My project was “Customer Churn Prediction using Machine Learning” focuses on analysing customer data to predict whether a customer is likely to leave a service. The objective of the project is to help businesses identify potential churn customers and take proactive measures to improve customer retention.

In this project, various data science techniques were applied, including data preprocessing, exploratory data analysis (EDA), and feature engineering. Machine learning models such as Logistic Regression, Decision Tree, and Random Forest were implemented and compared to evaluate their performance.

The models were trained and tested using customer-related features like tenure, monthly charges, and total charges. Performance was evaluated using metrics such as accuracy, confusion matrix, and classification report. Visualizations like histograms and correlation heatmaps were also used to understand data patterns.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

During the four weeks of this internship, I gained practical exposure to Data Science and Machine Learning concepts. I learned Python fundamentals, performed exploratory data analysis, and implemented machine learning models such as Logistic Regression, Decision Tree, and Random Forest.

Internships play a crucial role in career development as they provide real-world experience and help bridge the gap between theoretical knowledge and practical application. They also enhance problem-solving skills and improve confidence in handling technical tasks.

The project focused on predicting customer churn using machine learning techniques. The aim was to analyse customer data and identify patterns that indicate whether a customer is likely to leave a service.

This opportunity provided by USC/UCT allowed me to work on a real-world problem and gain hands-on experience. The program was well-structured, starting with basic concepts and gradually progressing to advanced topics and project implementation.

During this internship in Data Science and Machine Learning with Upskill Campus, I gained valuable practical knowledge and hands-on experience in working with real-world data and machine learning models. The internship helped me strengthen my understanding of Python programming and essential libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

One of the key learnings was understanding the importance of data preprocessing and exploratory data analysis (EDA). I learned how to clean and prepare datasets, handle categorical variables, and visualize data to extract meaningful insights. Implementing machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest helped me understand how models are trained and evaluated.

Thanks to Divya G and Vishnu for helping me throughout the project

To my juniors and peers, I strongly recommend actively participating in such internships. Focus on understanding the concepts rather than just completing tasks. Try to implement projects on your own, explore additional resources, and practice consistently. Do not hesitate to ask questions or seek guidance whenever needed.

2 Introduction

In today's data-driven world, organizations increasingly rely on data analysis and machine learning techniques to make informed decisions. One of the major challenges faced by businesses is customer churn, which refers to customers discontinuing the use of a product or service. Predicting customer churn is essential for companies to retain customers, improve satisfaction, and maintain steady revenue growth. Machine learning plays a vital role in analysing historical data and identifying patterns that help in predicting future customer behaviour.

2.1 Importance of Customer Churn Prediction

Customer churn prediction is a critical aspect of business analytics, as retaining existing customers is often more cost-effective than acquiring new ones. By identifying customers who are likely to leave, organizations can take proactive measures such as offering personalized services, discounts, or improved customer support. This not only helps in reducing revenue loss but also enhances customer satisfaction and loyalty.

In this project, machine learning techniques are used to analyse customer data and predict churn based on various factors such as tenure, monthly charges, and total charges. This approach enables businesses to make data-driven decisions and improve their retention strategies.

2.2 Role of Machine Learning in Data Analysis

Machine learning provides powerful tools for analysing large volumes of data and extracting meaningful insights. It enables systems to learn from historical data and make predictions without explicit programming. In the context of this project, machine learning algorithms are used to build predictive models that classify customers as churn or non-churn.

Various algorithms such as Logistic Regression, Decision Tree, and Random Forest are implemented to compare their performance and effectiveness. These models help in understanding customer behaviour patterns and improving prediction accuracy. Overall, machine

learning enhances the ability to solve complex problems and supports efficient decision-making processes.

2.3 Objective

The primary objective of this project is to develop a machine learning model capable of predicting customer churn based on available data. The project aims to analyse customer behaviour, identify patterns, and apply appropriate machine learning algorithms to improve prediction accuracy. It also focuses on comparing different models such as Logistic Regression, Decision Tree, and Random Forest to evaluate their performance. Additionally, the project helps in gaining practical experience in data preprocessing, exploratory data analysis, and model evaluation techniques.

2.4 Reference

The successful completion of this project was supported by various learning resources and references. Official documentation of Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn was used to understand implementation details. Online tutorials, educational videos, and data science blogs were referred to for gaining conceptual clarity and practical knowledge. These resources played an important role in guiding the development of the project and enhancing understanding of machine learning techniques.

2.5 Glossary

Terms	Acronym
Churn	The process in which customers stop using a product or service.
EDA	Exploratory Data Analysis
Machine Learning Model	An algorithm used to learn patterns from data and make predictions.
Target Variable	The output variable that the model aims to predict (Churn in this project).

3 Problem Statement

Customer churn is one of the most significant challenges faced by organizations, especially in industries such as telecommunications, banking, and subscription-based services. Churn refers to the situation where customers stop using a company's product or service over a period of time. This leads to a direct loss of revenue and affects the overall growth and sustainability of the business. Acquiring new customers is often more expensive than retaining existing ones, making churn prediction a critical area of focus for companies.

Traditional methods of analysing customer behaviour are often manual, time-consuming, and lack predictive capabilities. These methods fail to identify patterns and trends in large datasets, making it difficult for organizations to take timely actions to retain customers. As a result, there is a need for an automated and efficient system that can analyse customer data and provide accurate predictions regarding customer churn.

The problem addressed in this project is to develop a machine learning-based solution that can predict whether a customer is likely to churn based on various factors such as tenure, monthly charges, and total charges. By analysing historical customer data, the model aims to identify key patterns and relationships that influence customer decisions.

The implementation of such a predictive model can help organizations take proactive measures, such as offering personalized services or incentives, to reduce churn rates. This not only improves customer retention but also enhances business performance. Therefore, the objective of this project is to build an effective and reliable churn prediction system using machine learning techniques.

4 Existing and Proposed solution

In the existing system, customer churn is analysed using traditional and manual approaches. Organizations generally rely on historical records, basic statistical analysis, or simple rule-based methods to identify customers who may discontinue their services. These methods are often time-consuming and require significant human effort. Moreover, they lack the ability to process large volumes of data efficiently.

Another major limitation of the existing system is the absence of predictive capabilities. It mainly focuses on analysing past data rather than predicting future outcomes. As a result, organizations are unable to identify potential churn customers in advance and take necessary preventive actions. This leads to loss of valuable customers and decreased business performance.

Additionally, manual analysis may not capture complex relationships between different factors influencing customer behaviour. This reduces the accuracy and reliability of the results, making it difficult for businesses to make informed decisions.

The proposed system utilizes machine learning techniques to overcome the limitations of the existing system. It provides an automated and efficient approach for predicting customer churn based on historical data. The system begins with data preprocessing, where the dataset is cleaned and transformed into a suitable format for analysis.

After preprocessing, exploratory data analysis is performed to understand patterns and relationships within the data. Machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest are then implemented to build predictive models. These models are trained using the dataset and evaluated using performance metrics such as accuracy, confusion matrix, and classification report.

The proposed system offers several advantages, including improved accuracy, faster processing, and the ability to handle large datasets. It enables organizations to predict customer churn in advance and take proactive measures such as offering personalized services or incentives to retain customers.

4.1 Code submission (Github link):

https://github.com/gopireddyharshateja123/upskillCampus/blob/main/CustomerChurnPrediction.py

4.2 Report submission (Github link):

https://github.com/gopireddyharshateja123/upskillCampus/blob/main/CustomerChurnPredictionReport_Harsha_USC_UCT.pdf

5 Proposed Design/ Model

The proposed design of this project follows a structured machine learning workflow that transforms raw customer data into meaningful predictions about customer churn. The system is designed to process data through multiple stages, starting from data input and ending with the final prediction output. Each stage plays a crucial role in ensuring the accuracy and effectiveness of the model.

The process begins with the data input stage, where the dataset containing customer information such as gender, tenure, monthly charges, total charges, and churn status is collected. This raw data may contain inconsistencies or irrelevant information, which makes preprocessing an essential step.

In the data preprocessing stage, the dataset is cleaned and prepared for analysis. This includes handling missing values, removing unnecessary columns such as Customer ID, and converting categorical variables into numerical form using encoding techniques. This step ensures that the data is suitable for machine learning algorithms.

Next is the exploratory data analysis (EDA) stage, where statistical methods and visualizations such as histograms and heatmaps are used to understand the data distribution and relationships between features. This helps in identifying important patterns that influence customer churn.

After EDA, the process moves to the model building stage, where machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest are implemented. The dataset is divided into training and testing sets, and the models are trained using the training data.

In the evaluation stage, the performance of the models is measured using metrics such as accuracy, confusion matrix, and classification report. This helps in selecting the most effective model.

Finally, in the output stage, the trained model predicts whether a customer will churn or not. The overall design ensures a smooth flow from raw data to meaningful insights, making the system efficient and reliable for real-world applications.

5.1 High Level Diagram

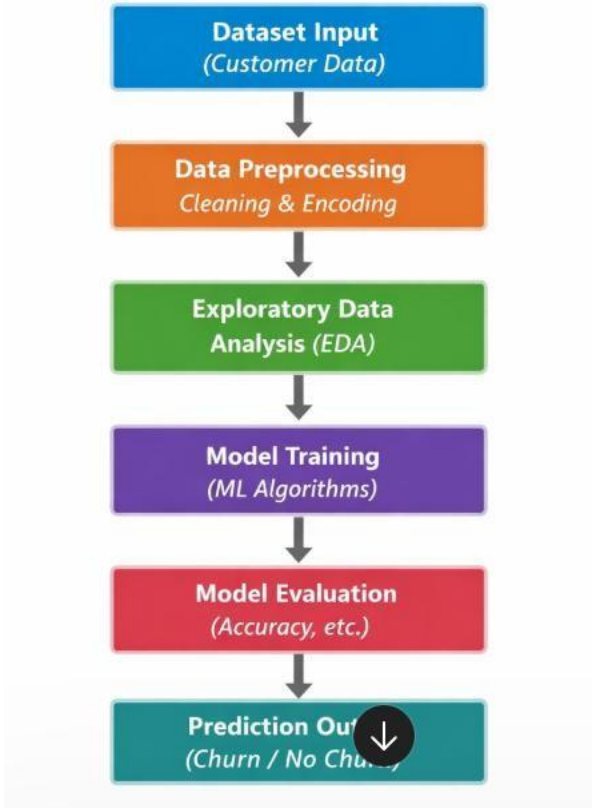


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

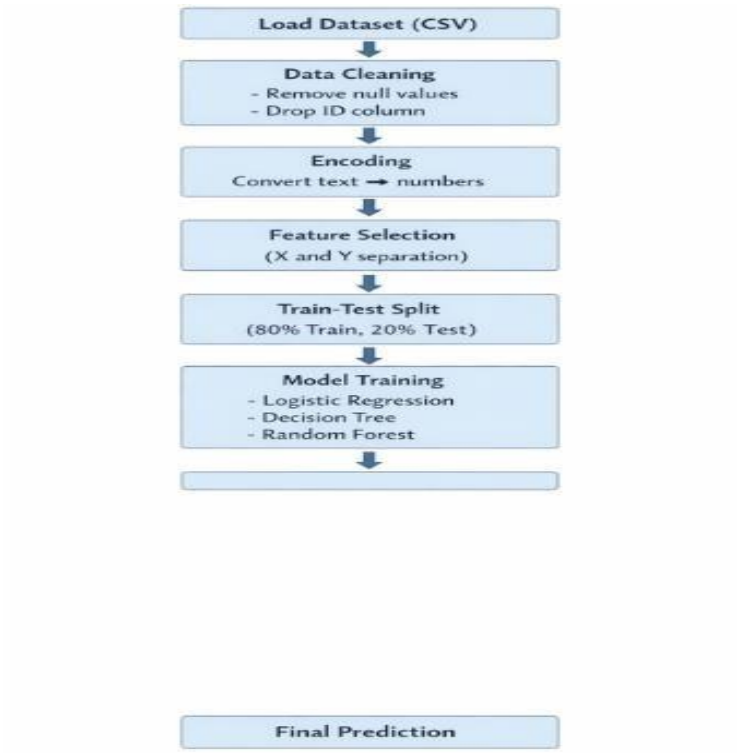
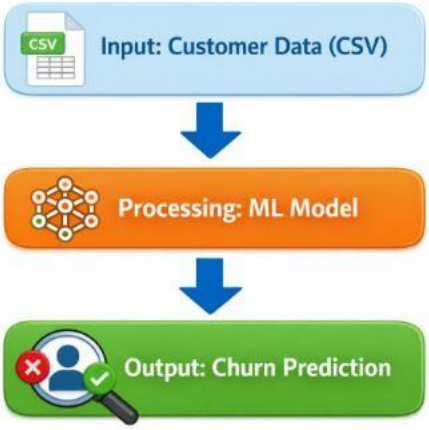


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces



6 Performance Test

The performance testing phase is a crucial part of this project as it demonstrates how the developed machine learning model can be applied in real-world industrial scenarios. Unlike a purely academic implementation, this section evaluates the system based on practical constraints such as accuracy, computational efficiency, memory usage, and scalability.

Several constraints were identified during the design of the customer churn prediction model. One of the primary constraints is accuracy**,** as the model must provide reliable predictions to be useful for business decision-making. Another constraint is data size, since the dataset used in this project is relatively small, which may limit the model's ability to generalize. Computation time (speed) is also important, as the system should process data efficiently without delays. Additionally, memory usage and scalability are critical factors when handling large datasets in real-world applications.

These constraints were addressed during the design phase by using efficient machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest. Data preprocessing techniques were applied to reduce unnecessary data and improve performance. The use of optimized libraries like Scikitlearn helped in improving computational efficiency.

Although some constraints like large-scale memory usage and high-speed processing could not be fully tested due to limited resources, their impact was analysed. It is recommended that in real-world deployment, larger datasets, optimized hardware, and advanced algorithms should be used to improve performance and scalability.

6.1 Test Plan/ Test Cases

The test plan was designed to evaluate the performance of the machine learning models under different conditions. The following test cases were considered:

- **Test Case 1:** Verify dataset loading and preprocessing
- **Test Case 2:** Check correctness of feature selection and encoding
- **Test Case 3:** Train models using training dataset
- **Test Case 4:** Test models using unseen data

- **Test Case 5:** Evaluate model performance using accuracy
- **Test Case 6:** Analyse confusion matrix and classification report

These test cases ensured that each stage of the system was functioning correctly and producing expected outputs.

6.2 Test Procedure

The following procedure was followed to test the system:

1. Load the dataset into the system
2. Perform data preprocessing (cleaning and encoding)
3. Split the dataset into training and testing sets
4. Train machine learning models using training data
5. Predict results using testing data
6. Evaluate model performance using accuracy, confusion matrix, and classification report
7. Compare results of different models

This systematic procedure ensured proper validation of the model.

6.3 Performance Outcome

The performance of the models was evaluated based on accuracy and other evaluation metrics. The results showed that:

- Logistic Regression provided consistent baseline performance
- Decision Tree captured patterns but may overfit
- Random Forest provided better robustness and reliability

Due to the small dataset, some limitations such as variation in results and warnings were observed. However, the models successfully demonstrated the ability to predict customer churn.

Overall, the system performed satisfactorily and proved that machine learning techniques can be effectively used for churn prediction. With larger datasets and further optimization, the model can be improved for real-world industrial applications.

7 My learnings

During the course of this internship, I gained valuable knowledge and hands-on experience in the field of Data Science and Machine Learning. This internship helped me understand how theoretical concepts learned in academics can be applied to solve real-world problems. One of the key learnings was the importance of data preprocessing and exploratory data analysis (EDA), which play a crucial role in improving the quality and performance of machine learning models.

I developed practical skills in using Python and its libraries such as Pandas, NumPy, Matplotlib, and Scikitlearn. I learned how to clean and transform data, perform feature selection, and implement machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest. Additionally, I gained experience in evaluating models using metrics like accuracy, confusion matrix, and classification report.

This internship also enhanced my problem-solving abilities, analytical thinking, and understanding of the complete machine learning workflow—from data collection to model deployment concepts. It helped me build confidence in handling real-world datasets and working independently on projects.

Overall, this experience will significantly contribute to my career growth by strengthening my technical foundation and preparing me for roles in Data Science and Machine Learning. It has also motivated me to further explore advanced concepts and improve my skills in this domain.

8 Future work scope

Although the current project successfully demonstrates the use of machine learning techniques for customer churn prediction, there are several areas for improvement and future enhancement. Due to time and resource limitations, certain advanced features and optimizations could not be implemented but can be explored in future work.

One of the major improvements would be the use of a larger and more diverse dataset, which would help in improving the accuracy and generalization of the model. A bigger dataset would also reduce issues such as overfitting and provide more reliable predictions. Additionally, more advanced machine learning algorithms such as Gradient Boosting, XG Boost, and Support Vector Machines can be implemented to achieve better performance.

Another area of enhancement is hyperparameter tuning, which can optimize the performance of models by selecting the best parameters. Feature engineering techniques can also be applied to create new meaningful features that may improve prediction accuracy.

Furthermore, the project can be extended by deploying the model as a web application or API, allowing real-time predictions in a practical environment. Integration with visualization dashboards can help businesses better understand customer behaviour and make data-driven decisions.

Overall, this project can be further developed into a scalable and industry-ready solution by incorporating advanced techniques and real-world deployment strategies.